

# scripts/autonomous-qa/lib-subsets.sh — User Guide

**Last verified:** 2026-07-26 (LVA-018 script-docs backfill cycle)

**Inheritance:** HelixConstitution §11.4.18 (script docs), §11.4.128 (evidence layout) **Classification:** project-specific

## Overview

Sourceable library that emits every **non-empty subset** of the matrix providers as the autonomous-QA provider-mix matrix. Operator decision (2026-06-30): broaden to ALL backed providers =  $\{\text{client-onboardable}\} \cap \{\text{API-backed}\}$  — **EXHAUSTIVE**, all 31 subsets  $\times$  backends  $\times$  queries.

Plan: docs/superpowers/plans/2026-06-29-autonomous-qa-backend-provider-matrix.md.

## The provider set

```
QA_PROVIDERS=(rutracker nmclub kinozal archiveorg gutenber)
```

The ids **MUST** match `ProviderSpec.forId()` in Challenge70 + the tracker descriptors. The set is the intersection of:

- **client-onboardable** — providers with a native bundled descriptor under `core/tracker/*` (in the production onboarding flow the picker is repopulated from the chosen API's catalogue, which REPLACES the bundled set — so a native provider is onboardable iff the API also vends it), and
- **API-backed** — the ids the live lava-api-go `/providers` endpoint vends.

The intersection **EXCLUDES** `rutor` (native but NOT in `/providers`) and `iptorrents` (Jackett-only, enabled separately via `lib-jackett.sh`).

## Output format

`qa_emit_subsets` prints all  $2^n - 1$  non-empty subsets, one per line:

<csv>|<slug>|<statehash>

- csv — comma-joined provider ids (passed to Challenge70 qa\_providers)
- slug — dash-joined provider ids (evidence dir name)
- statehash — short sha1 of the csv (state-hash per the §11.4.128 layout)

## Usage

```
# As a library (what run-matrix.sh does)
source scripts/autonomous-qa/lib-subsets.sh
mapfile -t SUBSET_LINES < <(qa_emit_subsets)
```

```
# Directly, to inspect the matrix (31 lines expected)
bash scripts/autonomous-qa/lib-subsets.sh
```

## Exit codes

- 0 — always (pure enumeration; no failure modes)

## Companion files

- scripts/autonomous-qa/run-matrix.sh — the consumer
- app/src/androidTest/kotlin/lava/app/challenges/ — Challenge70 provider matrix test